

Yatin Kande

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Summary

Data Scientist and ML Engineer with hands-on experience building and deploying production ML systems across computer vision, Generative AI, and predictive modeling. Delivered 84% RAG retrieval precision, 81% AUC-ROC churn detection, and 40% inference latency reduction on AWS, converting business problems into measurable AI outcomes across automotive, agricultural, and technology.

Education

University of Michigan Dearborn

M.S. in Data Science

Dearborn, MI

Aug 2024 – Dec 2025 | GPA: 3.77

Jain University

B.S. (Honors) in Data Science

Bangalore, India

Jul 2020 – May 2023 | GPA: 3.6

Technical Skills

- **Languages & Libraries:** Python, SQL, R, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, TensorFlow
- **ML & AI:** Regression, Classification, Clustering, Feature Engineering, RAG, LLM Fine-Tuning, Computer Vision, NLP
- **Cloud & Infrastructure:** AWS (S3, Lambda, SageMaker, DynamoDB, Lex), Docker, Kubernetes, Apache Spark, MLflow
- **Visualization & Tools:** Tableau, Power BI, Streamlit, Git, CI/CD

Professional Experience

Data Analyst Intern (AI/ML)

Feb 2024 – Jul 2024

DataZymes

Bengaluru, India

- Deployed **YOLOv5** crop disease detection on 10,000+ field images achieving mAP 74% and F1 0.76, enabling early farm intervention decisions.
- Cut inference latency by **40%** via batched processing and edge-compute optimization; reduced preprocessing time **3–4x** using Apache Spark + OpenCV, unblocking real-time field deployment.
- Boosted minority-class F1 from **0.72 to 0.76** with 1,500+ synthetic images, directly expanding rare disease detection coverage across underrepresented crop conditions.
- Tracked precision and recall across **5 model iterations** via Power BI dashboards, surfacing actionable insights that cut weekly engineering review time and accelerated release cadence.

Machine Learning Intern

Mar 2022 – Jun 2022

SmartKnower

Bengaluru, India

- Built an **XGBoost** churn model on 8,500+ records achieving 81% AUC-ROC, enabling sales teams to prioritize at-risk accounts for targeted retention outreach.
- Benchmarked **Random Forest, Logistic Regression, and XGBoost** via stratified cross-validation, selecting the optimal model based on precision-recall tradeoff — reducing false positives in high-stakes retention decisions.
- Engineered **15+ behavioral features** from temporal data, lifting baseline accuracy by 5 pp; applied SMOTE oversampling to improve minority-class detection by 6%.
- Gained **4% precision** via GridSearchCV tuning while sustaining 75% recall, increasing confidence in high-value retention interventions.

Projects

AutoRAG: Automotive Diagnostic AI Assistant

Nov 2025 – Jan 2026

- Achieved **84% retrieval precision** on automotive repair queries via hybrid RAG (FAISS + BM25) over PDF manuals and OBD-II images, reducing mechanic diagnostic lookup time.
- Boosted answer quality **12% over baseline** using Gemini 2.0 Flash; integrated Llama 3.2 vision as offline fallback ensuring zero downtime in low-connectivity environments.

Dataset Recommendation Chatbot (AWS Serverless)

Aug 2025 – Oct 2025

- Reduced dataset discovery to **under 10 seconds** via serverless AWS Lex + Lambda chatbot surfacing Kaggle and HuggingFace results with history-aware DynamoDB recommendations.
- Enabled **async processing at scale** through S3 + SQS + SNS event-driven architecture, decoupling ingestion from delivery with real-time email notifications.

Visual Speech Recognition with Deep Learning (LipNet)

Apr 2025 – Jul 2025

- Achieved **71% word accuracy and 19% CER** on the GRID corpus using a 3D CNN-BiLSTM + CTC model, demonstrating viability for accessibility and silent-speech applications.
- Processed **28,000+ video frames** via automated TensorFlow pipeline; reduced overfitting by 8% through temporal jitter and frame-rate augmentation, improving generalization to unseen speakers.

Certifications

- Oracle Cloud Infrastructure (OCI) AI Foundations Associate (Oracle, 2025)
- Google Data Analytics Professional Certificate (Google, 2024)